

RadiX Dose

v2.3

Build an example clinic & run your first calculation.

This guide walks you through RadiX Dose from an empty install: first you'll **build a small example clinic** (a facility, a machine, a chamber and a beam), then you'll **run a reference-dosimetry calculation** with it. The example data used here is entirely fictional — use it to learn the workflow, then replace it with your own.

A note on the data

"General Hospital / Radiation Oncology" is a neutral example only. Never store patient-identifiable information in the app — it is built for quality-assurance data.

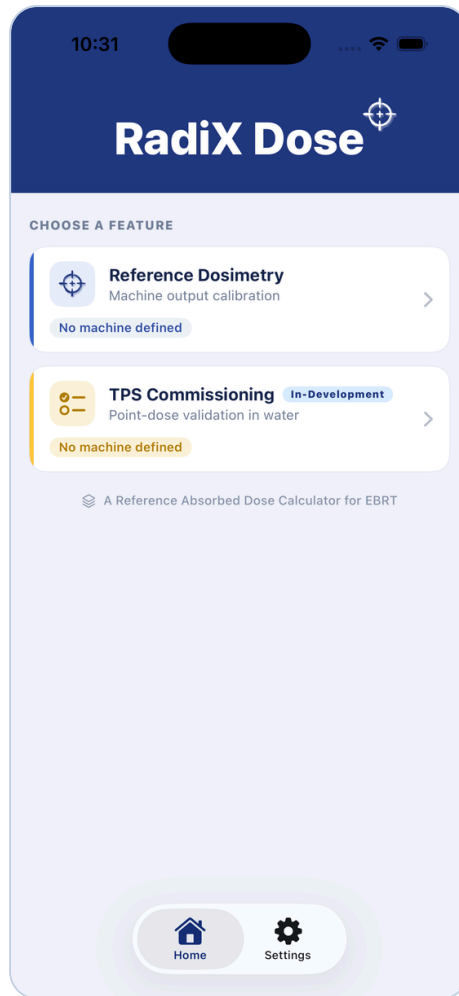
01 Open the app

1 Launch RadiX Dose and accept the agreement.

On first launch the **User Agreement & Disclaimer** appears. Read it, tick the box, and tap **Accept** (this is asked once).

2 Choose a feature.

On the home screen, tap **Reference Dosimetry** ("Machine output calibration"). The second feature, *TPS Commissioning*, is a preview and not needed here.



The home screen — tap Reference Dosimetry to begin.

02 Build the example clinic

Everything a calculation needs is defined once, in the **Settings** tab (bottom-right), as a nested library: **Facility → Machine → Detectors · Electrometers · Beams**. One facility and one machine are *active* at a time.

1 Add a facility.

Go to **Settings**. Under **FACILITY**, tap **+ Add**, then fill in:

- **Hospital Name** — e.g. General Hospital
- **Department Name** — e.g. Radiation Oncology

Tap **Add**. The facility appears and becomes **ACTIVE**.

10:29

Cancel New Facility Add

HOSPITAL NAME
e.g. Izmir Tinaztepe Galen Hospital

DEPARTMENT NAME
e.g. Radiation Oncology Department

New Facility — hospital and department names.

2 Add a machine (linac).

Tap the **chevron** to the left of your facility to expand it, then tap **+ Add Machine** and set:

- **Manufacturer & Model** — e.g. Varian · TrueBeam
- **Machine ID / Serial** — e.g. LINAC-1
- **Reference Absorbed Dose Protocol** — IAEA TRS-398 Rev.1 (2024), IAEA TRS-398 (2006), or AAPM TG-51. This chooses which beam-quality index and k_Q data the app uses.
- **Setup Configuration** — SSD- or SAD-based
- **Dose Reporting Point** — report at $d_{\max} / z_{\text{ref}}$, or at the reference (calibration) depth

Tap **Add**. The machine becomes active.

10:27

Cancel New Machine Add

MANUFACTURER

Select manufacturer...

MACHINE ID / SERIAL

e.g. H1912345

REFERENCE ABSORBED DOSE PROTOCOL

IAEA TRS 398 Rev.1 (2024)

SETUP CONFIGURATION

SSD-Based Setup

Source-to-Surface Distance

DOSE REPORTING POINT

Dose at D_{max} — based on PDD_{10} (%)

Reference-depth dose scaled to d_{max} using PDD_{10} per energy.

10:28

CONFIGURATION SETTINGS

Facility Machine Detector Beam Libraries Backup Management

Lock a record to protect it - Import or export your settings

FACILITY + Add

General Hospital

Radiation Oncology

ACTIVE

+ Add Machine

BACKUP & RESTORE

Export everything — facilities, machines, detectors (with certificates), electrometers, beams (with factors such as K_{QA} , K_Q , $K_{Q_{ref}}$, etc.) and saved reports — to one file you can move to another phone, then Import it there.

Export Backup Import Backup

LEGAL

User Agreement & Disclaimer

Clinical-use responsibility & verification terms

RadiX Dose - Version 2.2 (4)

Developer & licence holder: Oğuzhan Ayrançoğlu

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Home Output Factors Results Settings

New Machine — pick the code of practice. The expanded facility with “Add Machine”.

3

Add a detector (ionization chamber).

Open the machine's **Detector library** and add a chamber:

- Pick a model from the built-in catalog (e.g. an IBA FC65-G Farmer) or define your own
- Enter its **serial number** and **cavity volume / length**
- Enter the $N_{D,w}$ calibration certificate — value, date, and validity period — and choose the ^{60}Co or **cross-calibration** route

Save. Detector cards warn you as the certificate nears expiry.

14:17

Cancel **New Detector** Add

2.31 cm

Auto-filled from the detector library (vendor brochure). Tap the pencil to override.

BEAM CALIBRATION ROUTE ($N_{D,w}$)

^{60}Co Calibration

SERIAL NUMBER

1001

DOSE-TO-WATER CALIBRATION(S) — $N_{D,w}$ PER ELECTROMETER

ELECTROMETER

IBA — DOSE-1 · S/N: 2001

CALIBRATION FACTOR $N_{D,w}$ (Gy / nC)

0.04776

CALIBRATION VOLTAGE

+ - e.g. 300 V

TRACK CALIBRATION EXPIRY

CALIBRATION CERTIFICATE, ISSUING LABORATORY

PSDL, SSDL or Other...

Add PDF Certificate

PDF only · File size can up to 8,4 MB

+ Add electrometer & $N_{D,w}$

New Detector — chamber, serial, and the $N_{D,w}$ calibration per electrometer.

4

Add an electrometer (optional).

In the **Electrometer library**, add make/model/serial and, if applicable, a calibration factor k_{elec} (default 1.000).

11:06

Cancel New Electrometer Add

MANUFACTURER

IBA

MODEL

DOSE-1

SERIAL NUMBER

2001

ELECTROMETER CALIBRATION FACTOR (K_{Elec})

1.000

i Under IAEA TRS-398, record $K_{Elec} = 1.000$ when the ionization chamber and electrometer are calibrated together, since the combined $N_{p,w}$ already contains the electrometer response. Enter a separate K_{Elec} from the electrometer's own calibration certificate (in nC/rdg, or a dimensionless value near unity for a charge-reading electrometer) only when the chamber and electrometer are calibrated separately.

New Electrometer — make, model, serial, and k_{elec} .

5

Add a beam.

In the **Beam library**, add a **Photon** (or Electron) beam:

- **Energy** — e.g. 6 MV
- The **beam-quality index** your protocol needs — $\text{TPR}_{20,10}$, $\%dd(10)_x$, or R_{50}
- **Calibration conditions** — field size, delivered MU, and expected dose (used for the output check), plus PDD/TMR at the reference depth

Save.

14:21

Cancel New Photon Beam Add

ENERGY (MV)

6 FFF

DOSE MAXIMUM (D_{Max}) (g/cm^2)

1.5

APPROX. D_{Max} (g/cm^3)

1.50

PERCENTAGE DEPTH DOSE at 10cm (PDD_{10}) (%)

66.5

TISSUE-PHANTOM RATIO 20cm / 10cm ($\text{TPR}_{20/10}$)

0.67

New Photon Beam — energy, PDD_{10} , and $\text{TPR}_{20,10}$.

DEMO VALUES — REPRODUCE THE WORKED EXAMPLE

Enter these to reproduce the calculation shown in the next section, then adjust any of them to explore the app.

Machine	Varian TrueBeam · ID LINAC-1 · IAEA TRS-398 (2006) · SSD-based · reference depth 10 cm · 10×10 cm · 100 MU → 100 cGy expected
Electrometer	IBA DOSE-1 · S/N 2001 · k_{elec} 1.000
Detector	IBA FC65-G Farmer · S/N 1001 · 0.65 cm ³ · cavity 2.31 cm · $N_{D,w}$ 0.04776 Gy/nC · ⁶⁰ Co route
Photon beam	6 MV · $\text{TPR}_{20,10}$ 0.670 · PDD_{10} 66.5% · d_{max} 1.5 cm
Conditions & reading	20.0 °C · 1013.3 hPa → k_{TP} 1.000 · M = 14.02 nC (single reading)
Correction factors	k_Q 0.9957 · k_{pol} 1.0010 (M_+ 10, M_- 10.02) · k_s 1.0030 (M_1 10.0000, M_2 9.9390, V_1/V_2 3, Pulsed) · k_{pw} 1.0000 · k_{elec} 1.0000
Result	≈ 100.6 cGy at d_{max} (100.648) · +0.65 % vs 100 cGy expected

Note: the app requires each detector and beam to be **locked** (Settings ▶ Detector / Beam Library) before it computes a dose, and the $k_{\text{pol}}/k_s/k_{\text{pw}}$ factors are set on the **Factors** tab. Any value can be changed at any time.

⚡ Shortcut — load a ready-made example

To skip the manual setup, use **Settings → Import Backup** and select a sample backup file: it loads a complete demo clinic (facility, machine, chamber, beam, and saved records) in one step. Export Backup does the reverse — everything to one shareable file.

03 Run a calculation

Open the **Output** tab. Your active facility, machine, detector and beam are shown at the top.

1 Check the conditions.

Confirm **temperature** and **pressure** — the air-density correction k_{TP} is derived automatically.

2 Select the detector and beam.

Tap your chamber under *Detector Information*, and your energy under *Beam Information* (Photon / Electron).

3 Enter the charge reading.

Type **M₁** (add M₂/M₃ to average up to three). The **absorbed dose to water** and the **% deviation vs expected** appear live on the gauge.

4 Save or report.

Tap **Save Result** to record it, or **Report** to generate a PDF.

The screenshot shows a mobile app interface for calculating absorbed dose. At the top, the status bar shows the time 14:37 and battery level. The main section displays a large reading of 14.02 nC for M₁. Below this are input fields for M₂ and M₃. A note states: "Using a single reading · additional readings can be added to average". The calculated absorbed dose is 100.648 cGy at 1.5 cm (D_{Max}). The dose difference vs expected is 0.65%, which is within the ±2% tolerance. A graph shows a normal distribution curve with a green dot at 0.65%. At the bottom, there are buttons for "Save Result" and "Report". Below these are active correction factors based on IAEA TRS-398 (2006): K_Q = 0.9957, K_{Pol} = 1.0010, K_S = 1.0030, K_{P/W} = 1.0000, and K_{Elec} = 1.0000. The bottom navigation bar includes icons for Home, Output, Factors, Results, and Settings.

CHARGE READING 14:37

M₁ 14.02 nC C

M₂ — M₃ —

Using a single reading · additional readings can be added to average

CALCULATED ABSORBED DOSE

100.648 cGy

at 1.5 cm (D_{Max})

DOSE DIFFERENCE VS EXPECTED

0.65 %

Within ±2% tolerance

Save Result Report

ACTIVE CORRECTION FACTORS (Based on IAEA TRS-398 (2006))

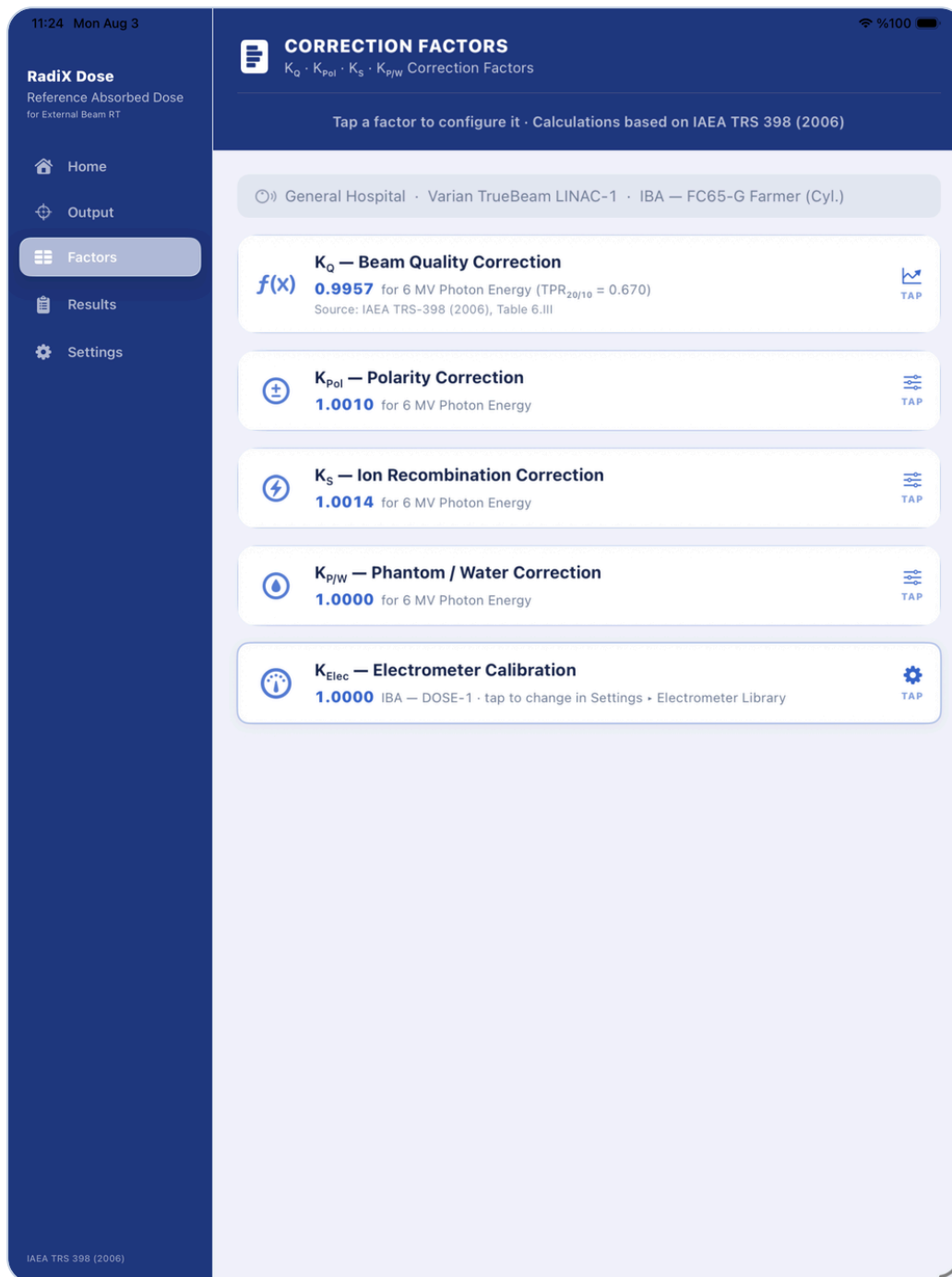
K _Q	0.9957	K _{Pol}	1.0010
K _S	1.0030	K _{P/W}	1.0000
K _{Elec}	1.0000		

Home Output Factors Results Settings

Output — the chamber reading becomes an absorbed dose (here 100.648 cGy, +0.65 % within tolerance), with the active correction factors listed below.

04 Correction factors

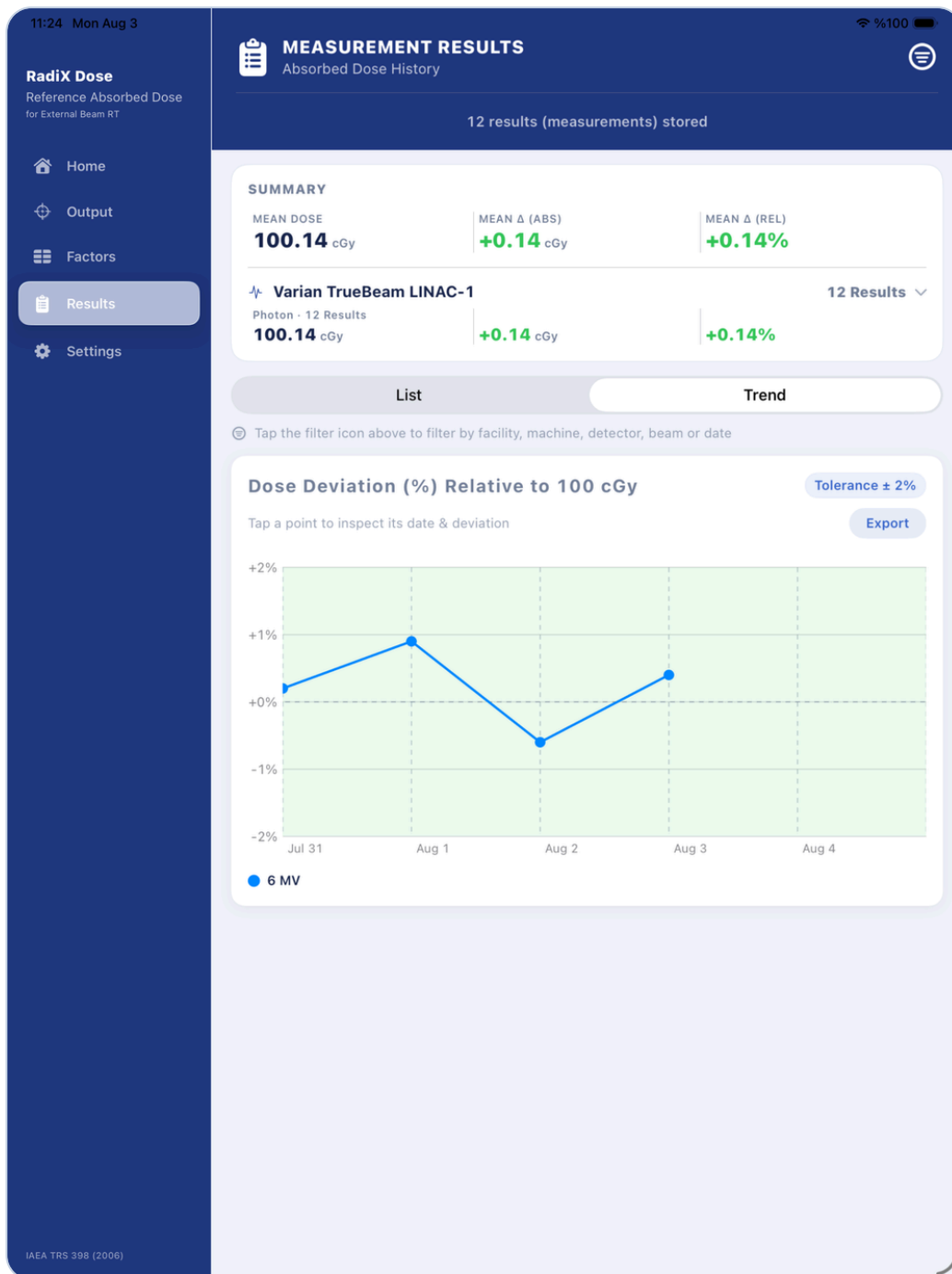
Open the **Factors** tab to measure and lock the corrections that aren't derived live — K_{Pol} (polarity), K_S (recombination), $K_{P/W}$ (phantom/water), and P_{gr} for TG-51 electrons. Each is stored **per chamber and energy**. A measured factor stays out of the dose until you **lock** it, so a half-entered value never silently changes a result.



Factors — every correction is shown and traceable; tap one to configure it.

05 Records & reporting

The **Results** tab keeps every saved calculation with its full factor snapshot — filterable by date and machine, with a scrolling trend chart so drift is visible at a glance. Use **Settings** → **Export Backup** to preserve or move your whole database.



Results — saved records and the output-trend chart.

To generate a report, tap **Report** in the **Output** view (next to *Save Result*). Tick the records to include, then **Preview & Export**.

Close

Reporting Records

✓

○

○

≡

✓

Aug 10, 2026 · 6 MV

100.65 cGy

Varian TrueBeam LINAC-1 · FC65-G Far...

+0.65%

1 of 1 selected

AP

Preview & Export

Reporting Records — select the measurements to include, then Preview & Export.

The result is a signed-off **A4 landscape PDF** — a trend chart above a full factor table (M , k_{TP} , k_Q , k_{Pol} , k_S , $k_{P/W}$, k_{Vol}), dose, deviation, and a Pass/Fail per record, with *Performed by* / *Reviewed by* sign-off lines — shareable or printable from the export button.

Reference Dosimetry Report

Tolerance ±2%

All measurements

Generated Aug 10, 2026 at 17:03 · 1 measurement · Page 1 of 1

+2%

+1%

+0%

-1%

-2%

Aug 8

Aug 9

Aug 10

Aug 11

● TrueBeam LINAC-16 MV

Date	Machine	Detector	Beam / Field	M (nC)	K_{TP}	K_Q	K_{Pol}	K_S	$K_{P/W}$	K_{Vol}	Dose (cGy)	$\Delta\%$	Result
Aug 10, 2026	TrueBeam LINAC-1	FC65-G Farmer... $N_{D,w}$: 0.04776	6 MV 10 × 10 cm	14.0200	1.0000	0.9957	1.0010	1.0030	1.0000	N/A	100.65 1.5 cm	+0.65	PASS

Performed by

Reviewed by

IAEA TRS 398 (2006) - absorbed dose to water - this report does not replace institutional QA procedures.

A4 · Landscape — the exported Reference Dosimetry Report (chart + full factor table, Pass/Fail per record).

Remember

RadiX Dose supports professionals — it never replaces the protocols, your judgement, or independent verification. Always verify every input, factor, and dose against the relevant code of practice before clinical use.

RadiX Dose v2.3 · Getting Started. A hands-on companion to the app — build an example clinic, then run your first reference-dosimetry calculation. © 2026 Oğuzhan Ayrancıoğlu (IAEA TRS-398 · AAPM TG-51).